MIL-D-80150B
2 November 1983
SUPERSEDING
MIL-D-80150A
10 April 1974

MILITARY SPECIFICATION

DRILLING MACHINES UPRIGHT, TURRET HEAD, SLIDING CARRIER, SIX SPINDLES, BENCH MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE. This specification covers electric motor driven, bench mounted, hand and power operated, six spindle, turret head, drilling, tapping, counterboring and reaming machines with equipment and accessories as described herein.

1.2 Classification. The drilling machines covered by this specification are of the following types, styles and sizes. The type, style and size supplied shall be as specified (see 6.2.1):

TYPE I - MANUAL OPERATION

Size	Drill Capacity	Swing
10	1/4 inch (6.35mm)	10 inch (254mm)
16	1/2 inch (12.7mm)	16 inch (406.40mm)
24	1/2 inch (12.7mm)	24 inch (609.60mm)
22	3/4 inch (19.05mm)	22 inch (558.80mm)

TYPE II - MANUAL AND AUTOMATIC OPERATION

Size	Drill Capacity	Swing
16	1/2 inch (12.7mm)	16 inch (406.40mm)
24	1/2 inch (12.7mm)	24 inch (609.60mm)
22	3/4 inch (19.05mm)	22 inch (558.80mm)

Style A - one 6-spindle unit
Style B - two 6-spindle units
Style C - three 6-spindle units

When other sizes are required the machine size and its corresponding characteristics shall be as specified (see Table I).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, Memphis, Tennessee 38114, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3413

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-18058 Machinery, Metal and Woodworking, Packaging of

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services

MILITARY

MIL-STD-461 - Electromagnetic Interference Control

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity of as directed by the contracting officer.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29CFR 1910)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B5.1 - T-Slots - Their Bolts, Nuts, Tongues and Cutters
B5.10 - Machine Tapers, Self-Holding and Strip Taper Series
B11.8 - Drilling, Milling and Boring Machines, Construction, Care and Use of Safety Requirements
ANSI/NFPA-79 - Electrical Standards for Metalworking Machine Tools
ANSI/IEEE-268 - Metric Practices (ASTM-380)

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(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY, 10018).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D3951 - Commercial Packaging, Standard Practice for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies).

2.3 Order of precedence. In the event of a conflict between the test of this specification and the references cited herein, the test of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a sample unit for first article inspection and approval (see 6.2.1 and 6.3).

3.2 Design. The drilling machine shall be new and one of the manufacturer's current models, capable of drilling workpieces within its specified capacity in accordance with the requirements herein. The machine shall include all components, parts and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage or distortion shall be accessible for adjustment, replacement and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in the U.S. Customary System of Units (US). These measurements may be converted to the International System of Units (SI) through the use of the conversion factors and methods specified in ANSI/IEEE268.

3.2.1.1 Measuring and indicating device calibrations. When specified (see 6.2.1), measuring and indicating devices such as feed screw dials, scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators and all other similar devices shall be graduated in the specified system (US or SI) of measurements. Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.2 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI Systems of Measurements. When a dual US and SI system is furnished, both feed dials shall have independent zero adjustments and both dials shall be calibrated in such a manner that the last dial graduation progresses and is continuous with the first dial graduation as the dial is rotated through the zero position.

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3.2.2 Reclaimed materials. The drilling machine may contain reclaimed materials provided such materials will not jeopardize the intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Controls. All operating controls shall be located conveniently to the operator at his normal work station.

3.2.4 Safety and health requirements. Covers, guards or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with operation of the machine. The safety devices shall prevent unintentional contact with the guarded part and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms and assemblies furnished on the machine, whether specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4). In addition, the machine shall comply with all requirements of ANSI B11.8 that are designated therein as the responsibility of the machine manufacturer. In the event of a conflict between the requirements of the OSHA and ANSI Standards, the requirements of OSHA shall apply.

3.2.5 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Each recirculating system shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings and filler caps shall be accessible.

3.2.6 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.3 Construction. The drilling machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No process such as peening, plugging or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Welding, brazing or soldering. Welding brazing or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts and similar fasteners which affect machine operation or prevent loss of fluids shall be installed with means of preventing loss of tightness. Fasteners that are subject to removal or adjustment shall not be swaged, peened or staked.

3.3.4 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scale, flux and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

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3.3.5 Painting. Unless otherwise specified (see 6.2.1), the drilling machine shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All dials that indicate stock removal or tool movement shall be graduated in increments not larger than 0.001 inch or .025mm. Dial diameters shall be such that graduations may be read from the normal operating position. Dials shall be permanently and legibly engraved or etched on a nonglare background.

3.3.8 Plates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background.

3.3.9 Gears. All gears shall be of a hardness and toughness that will enable the gear system to transmit full rated torque of the drive motor without gear damage, or premature wear.

3.4 Components. The machine shallconsist essentially of a base, column(s), sliding carrier, turret and spindle assembly, driving and indexing mechanisms, and operating controls.

3.4.1 Base. The base shall be internally ribbed and braced to withstand all loads which may be imposed during normal operations. The base shall form a foundation for the entire drilling machine. Each base shall be of the trough type. Working surfaces shall be accurately ground. A coolant trough shall be provided which shall extend completely around the periphery of the work surface with provisions for draining. When T-slots are provided, they shall conform to ANSI B5.1. Means shall be provided for securing the base to a bench, worktable, or cabinet. The base shall have means for accepting the column(s).

3.4.2 Column(s). The column shall be designed and proportioned to withstand all loads which may be imposed during normal operation of the machine. The column shall be arranged to accomodate a sliding carrier with turret and spindle assembly, speed and feed components, depth stop arrangement, and indexing and driving mechanisms. Means shall be provided to allow vertical travel of the sliding carrier. A graduated scale or other visible means shall provide an indication of height setting. The column shall be accurately fitted to the base.

3.4.3 Sliding Carrier. The sliding carrier shall be designed and proportioned to support and carry the driving and indexing mechanisms, turret slide, feed and speed components and depth stop arrangement. The carrier shall traverse vertically on the column and be adjustable over the full range of carrier travel. Means for locking after adjustment shall be provided. Carrier actuation shall be by manual means. Guideways shall be provided on the front of the carrier for supporting a turret slide. Gibs or other adjustable means shall be provided to compensate for guideway wear.

3.4.4 Turret slide. The turret slide shall be designed and proportioned to support and carry the speed, feed, and indexing mechanisms, depth stop arrangement and the turret and its components. The turret slide shall traverse vertically on guideways of the sliding carrier. Turret slide actuation shall be manually operated for Type I machines and shall have both manual and power

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operation for Type II machines. The turret slide face shall provide machined or scraped surfaces for accepting the indexing turret and spindle assembly.

3.4.4.1 Spindle gear box. The spindle gears shall be totally enclosed. Drive shafts and gearing shall be steel or steel alloy, heat treated and machined. Shafts shall be mounted in precision ball or roller bearings. Positive means for lubrication shall be provided to all applicable mechanisms enclosed within the gear box.

3.3.5 Turret. The turret shall be arranged with six equally spaced tool stations and be securely mounted on the turret slide face. The turret shall be carried in precision ball or roller bearings. An indexing mechanism shall be provided for selecting tool stations. Lubrication for spindle gears and bearings in operation shall be by oil bath. A sight gage shall be provided to indicate relative oil level.

3.4.5.1 Turret indexing. Turret indexing shall be by power. The index mechanism shall have a safety device to insure protection in the event of interference during normal indexing of the turret. A positive and accurate locating device shall lock the turret head in position when indexing is completed.

3.4.6 Spindles. Each spindle shall be of alloy steel, of one piece construction precision ground and mounted in ball or roller bearings. All spindles shall be individually locked in place throughout turret to prevent axial motion in the bearings. The machine shall be designed so that only the spindle to be used shall rotate. Each spindle shall have a threaded (male) spindle nose. Each spindle shall have its own individual speed and not less than two speed ranges, each spindle having a high and low speed in each range. The number of spindle speeds and their ranges shall be in accordance with Table I.

3.4.7 Main drive mechanism. The main drive mechanism shall be powered by an electric motor and an arrangement to provide motion to the turret and spindles. Spindle speeds shall be manually adjusted in either of two speed ranges. Spindle feeds shall be manual for Type I machines (by the use of a lever, spoked wheel, knob or other means) and automatic for Type II machines (by the use of a power feed mechanism).

3.4.8 Depth control. A quick-setting depth stop shall be provided for each spindle. Control shall be manual. Means shall be provided for maintaining adjustment to the tolerance test requirement in 4.5.3.

3.4.9 Styles B and C machines. Each Style B or C machine shall comprise a plurality of Style A machines as specified herein, mounted on a common base; two units for Style B machines and three units for Style C machines. Each unit shall operate independently and be interchangeable with the other(s). The unit center to center distances shall conform to Table I.

3.4.10 Type II machines. Each Type II machine shall be provided with both manual and power operation. Feeds shall be automatic and an arrangement to provide the following automatic cycle shall be incorporated in its design:

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After setup the operator shall be required only to load the workpiece, start the cycle and unload the workpiece. At the press of a button, the turret rapidly goes forward, tool automatically goes from rapid approach into feed. At end of stroke, or preset depth, the turret automatically retracts at any predetermined speed within its range until the tool clears the work, then into rapid retract. The turret shall automatically index and the next tool goes forward. This sequence shall continue until the operation is completed. The turret shall automatically stop to allow for fixture change or reloading a workpiece.

3.4.11 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to all requirements in this paragraph. The system shall conform to the requirements of ANSI/NFPA-79. Each machine shall draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit having a fused safety disconnect switch or circuit breaker. The drilling machine shall be initially wired for operation on 230 colts. An identified terminal for grounding the machine when installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in Table 240-A of ANSI/NFPA-79 for the disconnect fuse dripproof enclosure.

3.4.11.1 Motors. Motors shall have ball or roller bearings. Continuous duty horsepower ratings shall conform to the requirements of table I. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure

3.4.12 Coolant system. The coolant system, when furnished under 3.8(f), shall include a sump or reservoir, a power driven pump, and necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a strainer or baffle which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant.

3.4.13 Hydraulic system. When a hydraulic system is included in the machine design, the system shall be complete including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided on the high pressure line from the hydraulic pump to prevent damage to components. A filter system shall be provided to insure delivery of clean fluid to the hydraulic system. The hydraulic reservoir shall have a sight gage to indicate the fluid level.

3.5 Machine characteristics. The machine shall meet the applicable requirements of table I.

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TABLE I - SIZES AND CAPACITIES

CHARACTERISTIC	REQUIREMENTS						
	Type I				Type II		
	Size 10	Size 16	Size 22	Size 24	Size 16	Size 22	Size 24
Capacity							
Drill, in mild steel, inches	1/4	1/2	3/4	1/2	1/2	3/4	1/2 or 3/4
Tap, in mild steel, inches	#10/32	1/2 NF	3/4	1/2 NF	1/2 NF	3/4	1/2 NF or 3/4
Swing (drills to center of circle, diameter), inches	10	16	22	24	16	22	24
Throat depth, inches	5	8-1/8	11-1/2	12-1/2	8-1/2	11-1/2	12-1/2
Sliding Carrier							
Travel, inches	12	19	15	26	19	15	26
Turret							
Number of spindles	6	6	6	6	6	6	6
Turret travel, inches	2-3/4	6	8	6	6	8	6
Turret traverse rate, inches/minute (Type II only)	—	—	—	—	120	—	120
Spindles							
Number of spindles	6	7	4	7	7	4	7
Speed ranges per spindle	2	4	4	4	4	4	4
Speed ranges, RPM (see 3.4.6)	350 to 3300 650 to 6200	85 to 675 170 to 1355 255 to 2005 510 to 4010	80 to 500 160 to 1000 225 to 1500 450 to 3000	85 to 675 170 to 1355 255 to 2005 510 to 4010	85 to 675 255 to 4010 225 to 2005 510 to 4010	80 to 500 160 to 100 225 to 1500 450 to 3000	85 to 675 170 to 1355 255 to 2005 510 to 4010
Number feeds per spindle (Type II only)	—	—	—	—	9	—	9
Feed range, inches/minute, infinitely variable (Type II only	—	—	—	—	1 to 50	—	1 to 50
Clearance spindle to table, inches	12	22	—	26	22	—	26
Base							
Working surface, length by width, inches							
Style A	8 x 12	12 x 17	—	21 x 24	12 x 17	—	21 x 24
Style B	—	17 x 64	—	17 x 90	17 x 64	—	17 x 90
Style C	—	17 x 124		17 x 136	17 x 124		17 x 136
Center to center distance between units Styles B and C (see 3.4.9)	38	38	—	38	38	—	38
Motor horsepower							
Main drive	1/3	1	—	1	1	—	1

Requirements are not less than those shown. Where a range is shown, the requirement is from the smaller figure or less to the larger figure or more.

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3.6 Performance. Each machine shall be capable of performing the test requirements of 4.5.2.

3.7 Alignment tolerances. Each machine shall meet the alignment tests of 4.5.3.

3.8 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

a. One set of hand tools normally furnished by the manufacturer.

b. One drill chuck, type and size as specified (see 6.2.1).

c. One wear plate installed in the teable.

d. One reversible tapping head.

e. Spindle extensions, the number of each size and the size range as specified (see 6.2.1).

f. One floor cabinet, complete with coolant system and worklight.

3.9 Optional equipment. Optional equipment and accessories shall be furnished as specified (see 6.2.1).

3.10 Electromagnetic interference control. When specified (see 6.2.1), equipment procured under this specification shall comply with the requirements of MIL-STD-461.

3.11 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be securely attached to the machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The following information shall be furnished on the chart or plate:

Points of application

Serivce interval

Type of lubricant

Viscosity

Military or Federal Specification (if available)

3.12 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion resistant metal nameplate identifying the machine shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each such caption is clear as to its identity.

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Nomenclature

Manufacturer's name

Manufacturer's model designation

Drill capacity _____ inch or mm

Power input (volts, total amps, phase, frequency)

Contract number or order number

National Stock Number or Plant Equipment Code

Date of manufacture

3.13 Technical data. Technical data shall be furnished as specified (see 6.2.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such action is deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First Article inspection. When First Article approval is required under 3.1, First Article inspection shall be performed. Unless otherwise specified (see 6.2.1), First Article inspection shall comprise the examination in 4.4 and all tests in 4.5. Failure to pass the examination or any test shall be cause for disapproval of First Article.

4.3 Quality conformance inspection. Each machine shall be subjected to quality conformance inspection prior to being offered for acceptance. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.4, the 4.5 through 4.5.3, and the inspection in 4.6. Failure of the item to pass any examination, test, or inspection shall be cause for rejection.

4.4 Examination. The machine and equipment shall be examined to determine compliance with the requirements in 3.2 through 3.5 and 3.8 through 3.15.

4.5 Tests.

4.5.1 Operational test. Each machine shall be operated for not less than 30 minutes in accordance with the supplier's standard operating procedures. All mechanical functions of the machine shall be tested during this period. There shall be no evidence of overloading, overheating or excessive vibration during these operations.

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4.5.2 Performance tests. Each machine shall be tested in the following manner: Under manual control for Type I machines and under automatic control (after tool insertion and depth setting) for Type II machines, each machine shall drill and tap a hole of the maximum rated size for the machine 0.750 inch deep in SAE 1040 annealed steel, drilling being at a speed and feed rate which applies not less than rated load to the spindle drive, motor, without stalling, overloading or evidence of any permanent distortion. The depth of the hole shall be 0.750 plus or minus 0.002 inch. Threads shall be checked and shall conform to the applicable requirements of Handbook H28, Class 3B fit.

4.5.3 Alignment tests. Each machine and its components shall be tested for comformance with the alignment and accuracy tolerances specified in Table II. The instruments and methods used in performing the tests shall be at the option of the supplier provided they accurately determine the tolerances of the machine.

TABLE II

Alignment tolerances and machine accuracies.

REQUIREMENT	PERMISSIBLE TOLERANCE	MACHINE SIZE
Base working surface flat	0.0008"/ft	ALL
Column square with base inplane of spindle	0.0004"/ft	ALL
Spindle radial runout	0.001"/ft 0.0005"/ft	Size 10 16, 22, 24
Spindle parallel with column guideways	0.0008"/ft	ALL
Spindle travel square with base in longitudinal and transverse planes	0.0008"/ft	ALL
Slots of base working surface parallel with base	0.0005"/ft	ALL
Turret index accuracy, spindle to spindle at spindle nose	+0.0007"/ft ±0.0005"/ft	Size 10 16, 22, 24
Repeat index one spindle to itself at spindle nose	±0.00035"/ft ±0002"/ft	Size 10 16, 22, 24
Depth control accuracy	±0.002"/ft	ALL

4.5.4 Electromagnetic interference control tests. Equipment requiring electomagnetic interference control shall be tested form compliance with 3.11.

4.6 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of Section 5.

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5. PACKAGING

5.1 Packaging, packing and markings. Unless otherwise specified, packaging, packing and marking shall be in accordance with ASTM Standard D 3951. When required, level A packaging and level A or B packing and marking shall be in accordance with the applicable requirements of MIL-W-18058. The required levels and special markings, shall be as specified (see 6.2.1).

6. NOTES

6.1 Intended use. Drilling machines covered by this specification are intended for use in drilling, tapping, reaming, and countersinking operations in general shop work where multiple hole sizes in a single workpiece are required.

6.2 Ordering data.

6.2.1 Acquisition documents. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, size, and style required (see 1.2).

c. When a first article is required for inspection and approval (see 3.1, 4.2, and 6.3).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI) state required system (see 3.2.1.1).

f. Dual calibrations (US and SI); if required, (see 3.2.1.2).

g. Specify additional safety and health requirements (see 3.2.4).

i. Electrical system, if different (see 3.4.11).

j. Type motor enclosure, if different (see 3.4.11.1).

k. Standard equipment, if different (see 3.8).

1. Optional equipment, if required; fully describe required optional equipment (see 3.9).

m. Electromagnetic interference control, if required (see 3.10).

n. Lubrication chart or plate, if different (see 3.11).

o. Nameplate, if different (see 3.12).

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p. First article inspection, if different (see 4.2).

q. Quality conformance inspection, if different (see 4.3).

r. Specific packaging, packing and special marking, if required (see 5.1). (Specify level A packaging, specify level A or B packing and specify special marking if required.)

6.2.2 Contract data requirements. Required technical data, such as operators manuals, parts lists, wiring diagrams, and other instructions for operation and maintenance as identified on a numbered DD Form 1664 should be specified on DD Form 1423 incorporated into the contract.

6.3 First article. When first article approval is required under 3.1, the solicitation should include complete information concerning examination, testing and approval.

6.4 Safety and health requirements. Paragraph 3.2.4 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, heat, etc. Since OSHA limits the total hazard level ofthese hazards in the environment (and does not limit the hazard level of individual machines in the environment) the procuring agency is advised to analyze the existing hazards levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. If specific point-of-operation guarding the guard is required, the requesting activity should specify the exact configuration of the guard required, as in most cases, the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1(g).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - AL
Navy - SH
Air Force - 84
DLA - GS

User activities:
Army - SM
Navy - MC

Preparing activity:
DLA - IP

Project Number:
3413-0129

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-80150B	Drilling Machines Upright, Turret Head, Sliding Carrier,Bench Mou

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
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	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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